# JAPAGEN: Efficient Few/Zero-shot Learning via Japanese Training Dataset Generation with LLM

Takuro Fujii[1,2], Satoru Katsumata[3]

[1]Yokohama National University, [2]Nomura Research Institute, Ltd, [3]Retrieva, Inc.

tkr.fujii.ynu@gmail.com

# High Operational Costs for LLM Inference

- LLMs can solve various tasks in zero-shot setting [Brown+'20, Kojima+'22].

  ➡ They come with operational costs due to huge parameters.

- In specific scenarios, using task-specific models can reduce operational costs.
  - It is necessary to solve the task in zero-shot setting with performance equivalent to that of an LLM.
  - But, to do so, a large amount of training data is required, leading to the high costs of data collection.

# Can SuperGen perform effectively in Japanese?

SuperGen [Meng+'22] :

- SuperGen generates supervision for training encoder models (e.g., BERT), outperforming few/zero-shot prompting, as well as few-shot fine-tuning methods.
- SuperGen helps reduce the costs of collecting supervised data and the operational costs.

## Problem

- Powerful LLMs are primarily trained on English texts with limited exposure to other languages.
- It is crucial to investigate the effectiveness of SuperGen in mid-resource and different type languages.

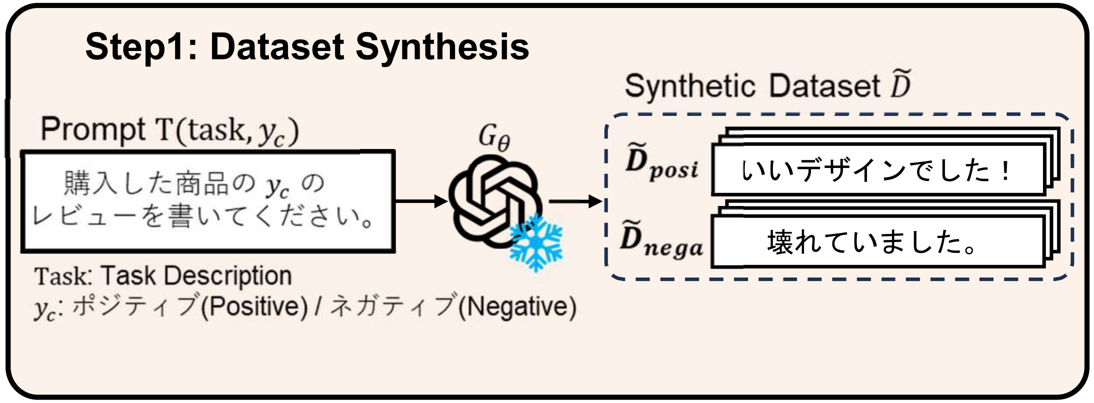
→ **We implement SuperGen in Japanese as a case study.**
(JapaGen)

# Overview of JAPAGEN (1/2)

**Step1:** Synthesis of supervised training data using LLM.

- Make a prompt $\mathrm{T}(\mathrm{task}, y_c)$ including task description $\mathrm{task}$ and a label $y_c$.
- LLM $G_\theta$ synthesizes input text $x_c$ corresponding to a label $y_c$.

**Step2:** Fine-Tuning small models $E_\phi$ (e.g., BERT) using synthetic data $\widetilde{D}$.



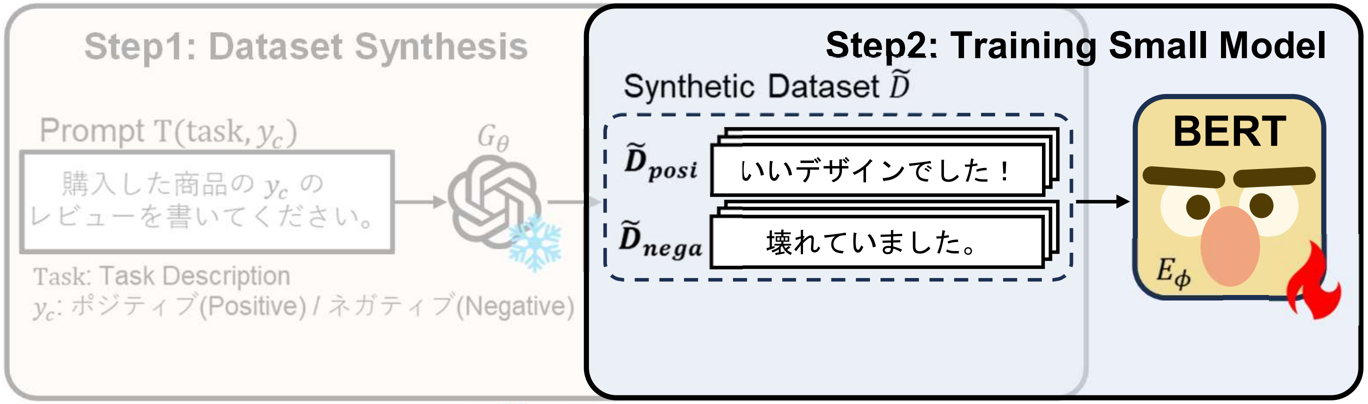

means model parameters are not updated.

# Overview of JapaGen (1/2)

**Step1:** Synthesis of supervised training data using LLM.

- Make a prompt $\mathrm{T}(\mathrm{task}, y_c)$ including task description $\mathrm{task}$ and a label $y_c$.
- LLM $G_\theta$ synthesizes input text $x_c$ corresponding to a label $y_c$.

**Step2:** Fine-Tuning small models $E_\phi$ (e.g., BERT) using synthetic data $\widetilde{D}$.

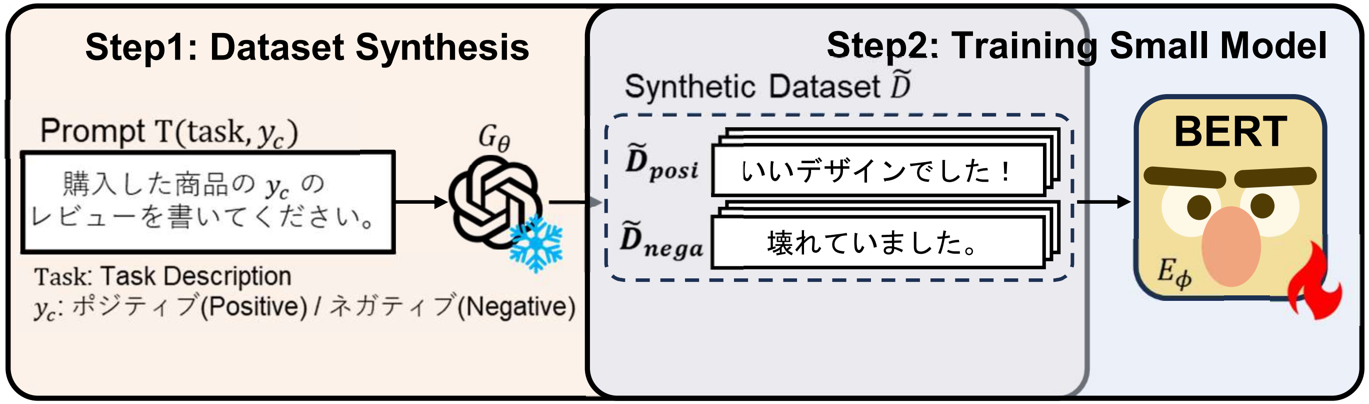



means model parameters are updated.

# Problem of SuperGen/JapaGen

## Low diversity of synthetic dataset

- Monotonous and low-informative prompt may limit the diversity of LLM output.
- Previous studies [Ye+'22] attempted to diversify synthetic text by adjusting hyperparameter (e.g., top-p, temperature).
  - However, we could not observe the same result in Japanese.




❄ means model parameters are not updated. 🔥 means model parameters are updated.

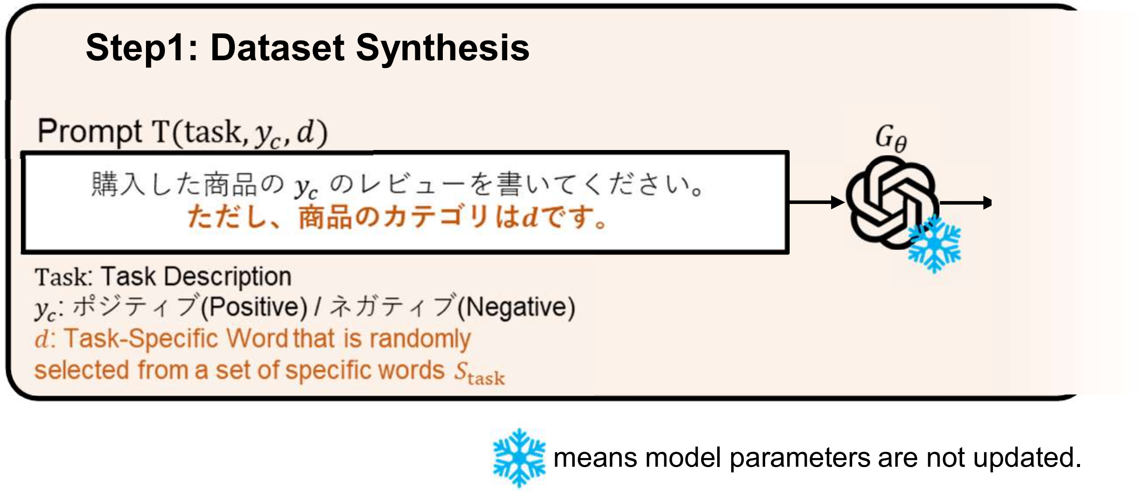
# Knowledge-Assisted Data Generation (KADG)

- To diversify synthetic dataset, we include a task-specific word $d$ in prompt.

  **Step1:** Manually create a set of task-specific words $S_{\text{task}}$.

  **Step2:** Randomly select a word from $S_{\text{task}}$, and include it in a prompt.

  ※ The subsequent process is the same as JAPAGEN.



# Experimental Setting

**Benchmarks:** six tasks including JGLUE [Kurihara'+22]

- MARC-ja, JCoLA, News, and COVID-19 (Single Text Classification)
- JNLI (Sentence Pair Classification)
- JSTS (Sentence Pair Similarity)

**Baselines:**

Setting 1 with <u>more parameters</u>

- Few / Zero-shot Prompting using GPT-3.5 turbo

Setting 2 with <u>more real dataset</u>

- Few-shot / Fully Supervised Fine-Tuning using pretrained BERT

# Experimental Setting

**Viewpoint:**

1. Downstream Performance
2. Distribution
3. Diversity & Label correctness
4. Data Scaling
5. Qualitative Evaluation

# Downstream Performance – *vs. Fine-Tuning* –

- JapaGen outperforms few-shot fine-tuning on five tasks.
  - Notably in JSTS, JapaGen achieves a Spearman score of 57.67%.

→ **JapaGen can be effective in the scenarios where the cost of data collection or annotation is high.**

| Method | MARC-ja Acc. | JSTS Spearman | JNLI Acc. | JCoLA Mcc. | News Acc. | COVID-19 Acc. | Avg. |
|---|---|---|---|---|---|---|---|
| **Fine-Tuning**: *fine-tuning pretrained BERT under gold data.* | | | | | | | |
| Fully Supervised | 95.78 | 87.47 | 90.19 | 40.62 | 95.75 | 78.49 | 82.82 |
| Few-Shot | 61.57 | 14.80 | 37.72 | -0.85 | 51.98 | 42.24 | 37.40 |
| **JapaGen**: *fine-tuning pretrained BERT under pseudo data generation via LLM.* | | | | | | | |
| Zero-Shot | **77.76** | **72.47** | **46.49** | **18.17** | **57.37** | 34.36 | **54.23** |
| Few-Shot | 62.97 | 72.56 | 50.82 | 14.54 | 62.86 | 43.13 | 51.15 |

# Downstream Performance – *vs. Prompting* –

- JAPAGEN achieves performance improvements on three tasks.
  - 3.94%, 4.96%, and 17.10% over zero-shot Prompting on JSTS, JNLI, News.

→ **JAPAGEN has the potential to outperform settings with more parameters and more annotated data.**

| **Method** | **MARC-ja** Acc. | **JSTS** Spearman | **JNLI** Acc. | **JCoLA** Mcc. | **News** Acc. | **COVID-19** Acc. | **Avg.** |
|---|---|---|---|---|---|---|---|
| **Prompting**: *prompt-based LLM learning.* | | | | | | | |
| Zero-Shot | 94.82 | 68.53 | 41.53 | 24.76 | 40.27 | 62.76 | 57.66 |
| Few-Shot | 97.38 | 78.50 | 35.86 | 26.00 | 44.82 | 65.44 | 61.72 |
| **JapaGen**: *fine-tuning pretrained BERT under pseudo data generation via LLM.* | | | | | | | |
| Zero-Shot | 77.76 | **72.47** | **46.49** | 18.17 | **57.37** | 34.36 | 54.23 |
| Few-Shot | 62.97 | 72.56 | 50.82 | 14.54 | 62.86 | 43.13 | 51.15 |

# Downstream Performance – *vs. KADG, Few-shot* –

- KADG outperforms zero-shot JapaGen only on MARC-ja and News.
- Few-shot JapaGen outperforms the zero-shot JapaGen on four tasks.
  - improvements of 4.33%, 5.49%, and 8.77% on JNLI, News, and COVID-19, respectively.

| Method | MARC-ja<br>Acc. | JSTS<br>Spearman | JNLI<br>Acc. | JCoLA<br>Mcc. | News<br>Acc. | COVID-19<br>Acc. | Avg. |
|---|---|---|---|---|---|---|---|
| **JapaGen**: *fine-tuning pretrained BERT under pseudo data generation via LLM.* | | | | | | | |
| Zero-Shot | 77.76 | 72.47 | 46.49 | 18.17 | 57.37 | 34.36 | 54.23 |
| w/ KADG | **83.24** | 71.49 | 46.04 | 16.22 | **59.00** | 26.29 | 50.38 |
| Few-Shot | 62.97 | **72.56** | **50.82** | 14.54 | **62.86** | **43.13** | **51.15** |

# Analytical Setting

**Viewpoints and Metrics:**

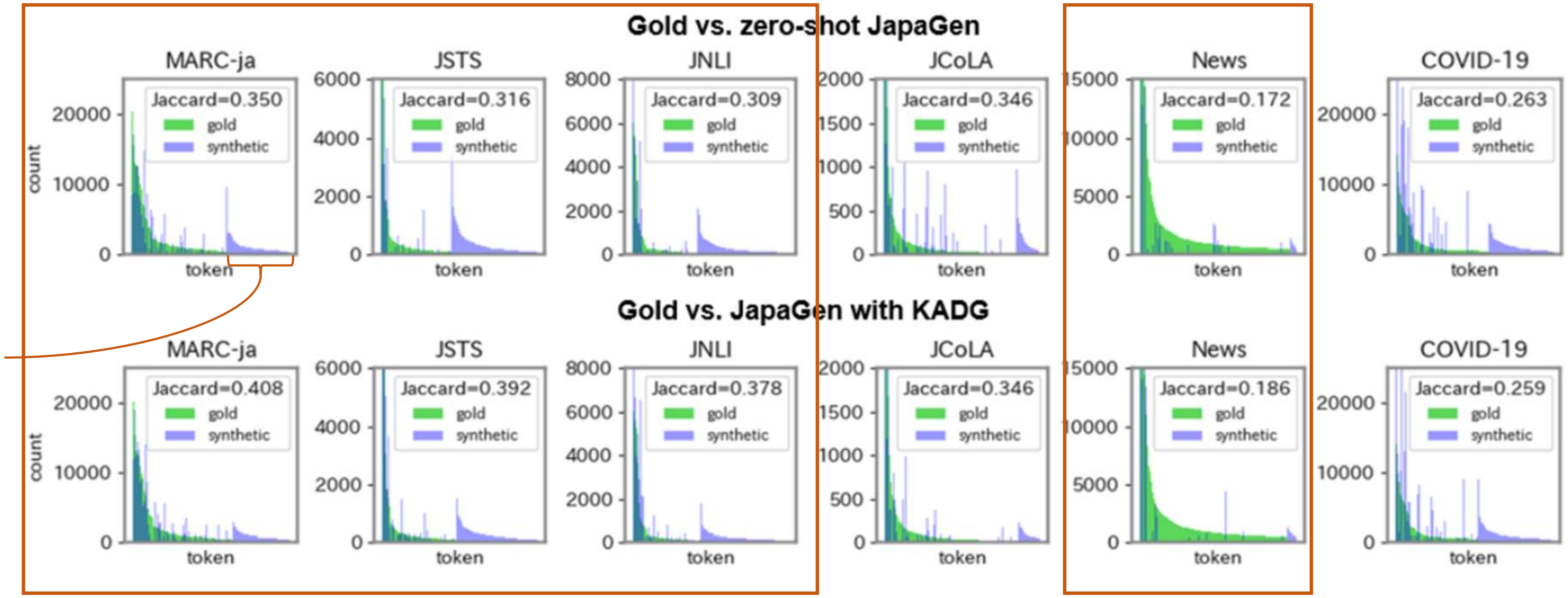
- **Distribution**
  - The degree of alignment between gold and synthetic dataset is calculated by Jaccard index based on the number token appearance.
- **Diversity**
  - Dataset diversity is calculated by Self-BLEU [Zhu+'18].
    Lower Self-BLEU score indicates higher dataset diversity.
- **Label Correctness**
  - To calculate label correctness, we train BERT on the gold dataset and then measure accuracy on the synthetic dataset.

# Analysis – Appeared Token Distribution – (1/2)

- KADG obtains a higher Jaccard index than zero-shot JAPAGEN on four tasks.
- Qualitatively, fewer words appear only in the synthetic data.

→ **The distribution of the data synthesized by KADG is more aligned with the gold data than that of zero-shot JAPAGEN.**

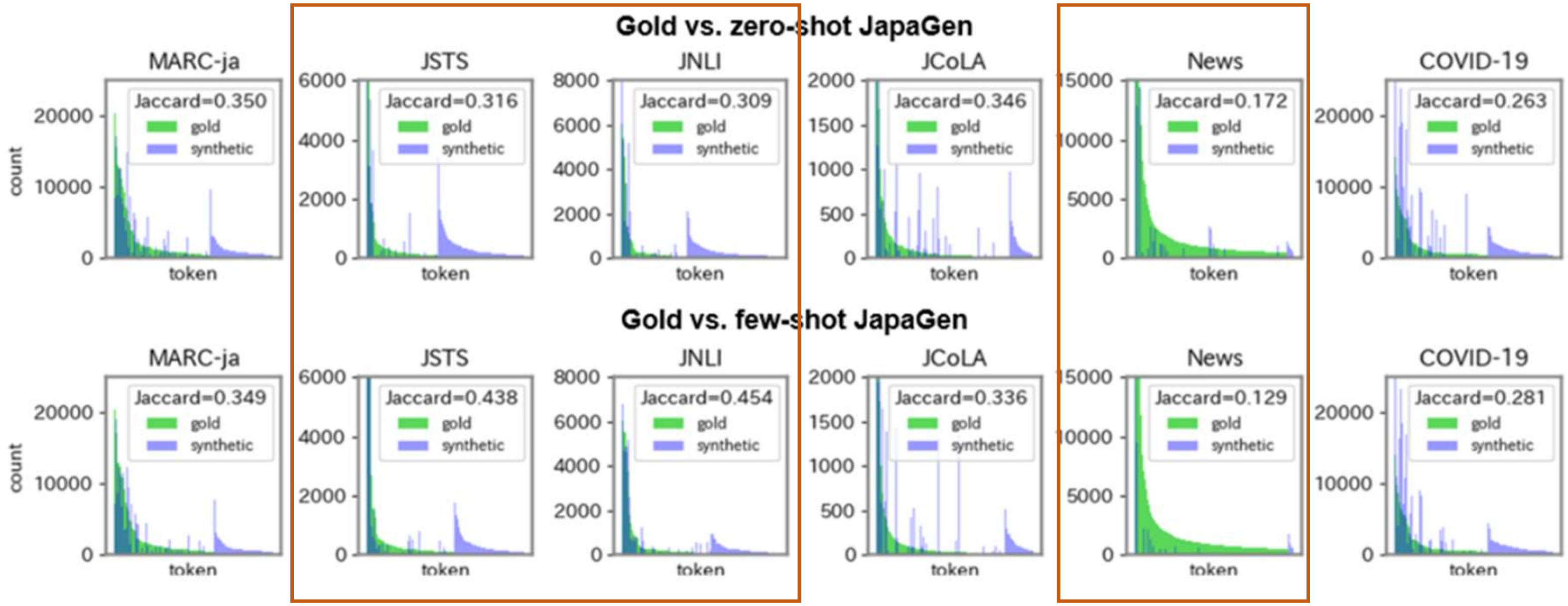
👀 These tokens appear only in the synthetic dataset.



# Analysis – Appeared Token Distribution – (2/2)

- Few-shot JAPAGEN obtains a higher Jaccard index than zero-shot on three tasks.
- Qualitatively, fewer words appear only in the synthetic data.

→ **The distribution of the data synthesized by few-shot JAPAGEN is more aligned with the gold data that of the zero-shot.**



# Analysis – Diversity and Label Correctness – (1/2)

- The trade-off between diversity and label correctness is observed as with the previous works.
  - Zero-shot JAPAGEN can synthesize datasets with a diversity similar to the gold dataset in JSTS, JNLI, and JCoLA.
  - The label correctness in those tasks is not as high as the gold dataset.

| Method | MARC-ja | JSTS | JNLI | JCoLA | News | COVID-19 |
|---|---|---|---|---|---|---|
| | | | ***Diversity*** (%) | | | |
| Gold | 40.53 | 72.93 | 72.94 | 56.66 | 62.97 | 43.14 |
| Zero-shot | 91.67 | **74.89** | **69.97** | **65.80** | 79.90 | 84.31 |
| | | | ***Label Correctness*** (%) | | | |
| Gold | 99.06 | 0.137 | 98.01 | 96.28 | 98.89 | 90.87 |
| Zero-shot | 99.97 | 1.540 | 35.11 | 66.34 | 49.84 | 60.80 |

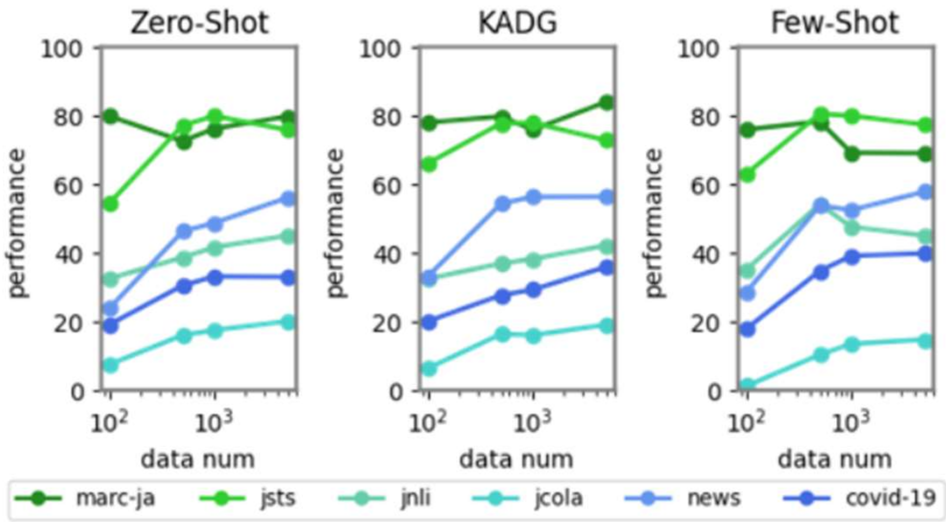
# Analysis – Diversity and Label Correctness – (2/2)

- Despite the trade-off, in MARC-ja, KADG improves the Self-BLEU score without compromising label correctness, enhancing downstream performance.
- Few-shot setting, in JSTS and JNLI, yield results similar to the zero-shot in in diversity, but we observe improvements in label correctness.

| Method | MARC-ja | JSTS | JNLI | JCoLA | News | COVID-19 |
|---|---|---|---|---|---|---|
| | | | ***Diversity*** (%) | | | |
| Zero-shot | 91.67 | 74.89 | 69.97 | 65.80 | 79.90 | 84.31 |
| w/ KADG | **84.97** | 76.12 | 73.13 | 78.91 | 82.93 | 81.91 |
| Few-shot | 90.25 | 81.80 | 78.28 | 67.15 | 79.25 | 83.40 |
| | | | ***Label Correctness*** (%) | | | |
| Zero-shot | 99.97 | 1.540 | 35.11 | 66.34 | 49.84 | 60.80 |
| w/ KADG | **99.96** | 1.540 | 39.37 | 63.94 | 43.61 | 58.86 |
| Few-shot | 99.90 | **1.094** | **50.16** | 63.33 | 57.33 | 64.43 |

# Analysis – Data Scaling –

- **Performances improve as the data size increases.**
  - Performance tends to saturate, as the perfomances with 5,000 samples are near to those with 50,000 samples.



# Analysis – Qualitative Evaluation – (1/2)

**Overall**: JAPAGEN generated texts aligned with labels except for JSTS and JCoLA.

**JSTS:**
A regression task to predict the semantic similarity score between two sentences.

**Result:**
While labels are continuous, discrete values as labels in the prompt limits the capability of JAPAGEN to capture detailed similarity between two sentences.

| Task | Synthesized Text | Label |
|---|---|---|
| JSTS | 子供たちが講演で楽しそうに遊んでいます。<br>(The children are having fun playing in the park.)<br>講演で遊ぶ子供たちが笑顔で何かを楽しんでいます。<br>(The children playing in the park are smiling and enjoying something.) | Similarity =1.0 |

# Analysis – Qualitative Evaluation – (2/2)

**Overall**: JAPAGEN generated texts aligned with labels except for JSTS and JCoLA.

**JCoLA:**
A binary classification task to predict whether a Japanese text is syntactically acceptable or unacceptable.

**Result:** LLM struggles with generating unacceptable sentences. LLM often generate not a syntactic error but a typo.

| Task | Synthesized Text | Label |
|---|---|---|
| JCoLA | 私は友達と昨日<u>**食べった**</u>寿司にします。<br>(I will have the sushi I <u>**atte**</u> with my friends yesterday.) | Unacceptable |

# JapaGen is effective for formal text classification

### What we do?

- We investigate the effectiveness of SuperGen in a mid-resource language with characteristics different from English, as a case study in Japanese.

### Result

- JapaGen is particularly effective for classification tasks where the input consists of formal text compared to few-shot Fine-Tuning and Promoting.

  ➞ The potential to surpass scenarios with more parameters and more annotations.
- The result of KADG and few-shot JapaGen indicates incorporating task knowledge and examples into the prompts can further enhance its capabilities.

### Challenge

- Low label correctness and the difficulty in synthesizing datasets with continuous value labels such as JSTS and with the desired grammatical errors in JCoLA.

# Thank you for listening!
# Feel free to ask!